VLM-Verified Counterfactual Hindsight for Sparse-Reward Manipulation*

Daniel Grant **Parshawn Gerafian** **Matei Gardea**
Department of Electrical Engineering and Computer Sciences
University of California, Berkeley
{dgrant,pgerafian,mgardea}@berkeley.edu

Extended Abstract

Problem. Sparse-reward manipulation is a credit-assignment problem. On Gymnasium-Robotics Fetch [3] (Push / PickAndPlace / Slide) a fresh policy succeeds on under 1% of episodes, and the single terminal reward bit at the end of a 50-step trajectory decays geometrically as it is bootstrapped backward through TD. HER relabels failed trajectories with realized achieved goals, but it never says *which* timestep made failure inevitable. Prioritized experience replay (PER) is also stuck: it sharpens toward high-TD-error transitions, and in the sparse regime the TD signal is exactly what is missing. Concurrent VLM-RB [22] adds a frozen vision-language model (VLM) score as an *additive* priority bias; we take the *multiplicative* form instead, and bolt on a simulator-verified counterfactual channel.

Methodological contributions. **(1) Semantic PER as IS-posterior reweighting.** Treat the VLM as an approximation $q_\phi(t^\star \,|\, \tau)$ to the posterior over the failure-causing timestep. Multiply the TD-error priority by q_ϕ. The standard PER weight remains a sound IS correction with a derivable bias bound; miscalibration shows up as variance inflation in μ, never as bias in the value target. The additive VLM-RB mixture silently retargets the Bellman loss to a λ-weighted objective it does not correct. **(2) VLM-Verified Counterfactual Hindsight.** Language-only counterfactuals have a dominant failure mode: 100% teleport-collapse on Opus 4.7 / FetchPush. Our fix asks the VLM for a *corrective action sequence*, executes it in a fork of the training simulator, and writes the trajectory to the buffer only if the env's own sparse reward fires. Confidence is then 1.0 and the modeling error is zero.

Headline findings. (i) **Verified-CF matches VLM-CF; wins on Slide.** After 500k SAC + HER + verified-CF updates verified-CF reaches mean success 0.606 across the three Fetch envs ($0.85/0.35/0.617$ on Push / PickAndPlace / Slide). VLM-CF reaches 0.622 — a tie within seed noise ($\Delta = -0.016$, $n{=}3$, $\pm$ standard error throughout). On Slide alone, verified-CF leads (0.617 vs. 0.550, $+0.067$). Both gain $+0.45/+0.434$ on Slide over PER@3M / HER@250k baselines (Fig. 1). (ii) **Pre-registered kill: honored.** The Phase-1 Oracle-CF ablation violated our $+0.10$ kill threshold on FetchPickAndPlace at the 250k decision horizon ($\Delta = -0.05$). We honored it and pivoted the headline to the IS-posterior framing and the verifier. (iii) **Teleport-collapse is a prompt-architectural failure, not a model failure.** On six synthetic and ten real Fetch failures we swept two vendors and three models (GPT-4o / Opus 4.7 / Sonnet 4.5): `(GPT-4o, achieved_goal)` teleports on $0/6$, Opus teleports on $4/4$ of the same configuration. (iv) **Cross-task transfer.** A single format-string prompt template yields $12/12$ parse / task-relevance / plausibility on Push / PickAndPlace / Slide.

Honest negative. Verified-CF has a cold-start failure mode. PnP seed 42 rejected the first ~ 80 VLM calls at 100%, so the agent ran as a more expensive HER for that window. The multiplicative-vs-additive distinction is analytical (Eq. 1); a direct VLM-RB head-to-head is pre-staged for camera-ready.

*CS 285 Final Project, Spring 2026. UC Berkeley, Department of EECS.

Preprint.

1 Introduction

Sparse-reward manipulation is a credit-assignment problem. A 50-step episode emits one reward bit at the terminal timestep, and TD-error decays geometrically as the signal bootstraps backward through the trajectory. Most transitions never see a usable gradient. On Gymnasium-Robotics Fetch the early policy succeeds on under 1% of episodes, so the replay buffer holds almost no positive signal to propagate backward in the first place. PER [21] sharpens sampling toward high-TD-error transitions, but the sharpening sits downstream of a credit signal that itself decays away from the rare goal-achievement timestep. Hindsight Experience Replay [1] fixes part of this by relabeling failures as successes against an achieved goal, but the relabel target is constrained to what the agent actually did. The one transition we most want flagged — the moment failure became inevitable — receives no special treatment.

Recent work plugs frozen VLMs into the RL loop as "oracles" for reward shaping [19, 10], replay-buffer prioritization [22], and failure detection in manipulation [5]. We read all of these as importance-sampled stochastic optimization with a foundation-model proposal distribution: the VLM is a frozen approximation to the (intractable) posterior $p(t^\star \mid \tau)$ over which transitions caused an episode's outcome. Four contributions follow.

(1) IS-posterior framing of VLM-guided replay (§3). We re-derive PER inside a non-uniform-replay taxonomy, placing our scheme as *proposal-shaping* with a frozen foundation-model credit-assignment oracle. The multiplicative form $\mu_P w_{\text{sem}}$ keeps PER's IS correction trajectory-conditionally well-defined. The additive mixture used by VLM-RB [22] retargets the loss to a λ-weighted objective and does not correct for the shift.

(2) VLM-Verified Counterfactual Hindsight (§3.3). A direct prompt for an alternative *achieved goal* fails: 100% of Claude Opus 4.7 calls on FetchPush return `corrective_position = desired_goal`, a degenerate zero-information output. Our fix is to ask for a corrective *action sequence* instead, then execute it in a fork of the training simulator. Teleporting a 50 g block by 50 cm with a 4-D end-effector action is physically impossible in MuJoCo [23], so the teleport mode cannot reach the buffer. This is a simulator-verified generator–verifier protocol (cf. 28).

(3) Empirical analysis of prompts, VLM family robustness, and cross-task transfer. A head-to-head GPT-4o vs. Claude Opus 4.7 sweep on six synthetic Fetch failures pins down a best non-degenerate configuration: `(GPT-4o, achieved_goal)` teleport-collapses on 0/6, against Opus's 4/4 of the same configuration. Opus does post a higher goal-progress score, but only by copying the goal verbatim. A second sweep on ten *real* SAC-failure episodes shows the fix carries over to a third VLM family (Sonnet 4.5). A 12-episode cross-task pilot lands 12/12 task-relevant annotations on Push / PickAndPlace / Slide with a single prompt template.

(4) Honest pre-registered negative result. A 36-run Oracle-CF kill experiment came in under our $+0.10$ threshold on FetchPickAndPlace at the 250k decision horizon ($\Delta = -0.05$ vs. HER). This caps the headroom for any VLM-in-replay method on Fetch at that horizon. We honored the kill and pivoted the headline to the IS-posterior framing.

2 Related Work

HER, PER, and counterfactual augmentation. Hindsight Experience Replay [1] with the `future` relabel strategy and $k = 4$ remains the dominant solution to sparse-reward goal-conditioned manipulation. Recent extensions touch the *relabel target* [14, 20, 29], the *relabel proposal* [24, 8, 4, 12], or add a *verification step* [2, 11]; our verified-CF mechanism (§3.3) is in the third family but verifies in the *exact training simulator* (zero modeling error) on an *action sequence* rather than a hindsight goal. Prioritized DQN [21] samples in proportion to $(|\delta| + \varepsilon)^\alpha$ and IS-corrects each gradient; the 2024–2026 PER lineage explores auxiliary multiplicative modifiers [18, 27, 25, 9].

Differentiation from VLM-RB [22]. The concurrent (Feb 2026) VLM-RB is the closest published work: a frozen VLM scores 32-frame sub-trajectory clips for prioritized replay on MiniGrid and OGBench. Two methodological differences matter here. **First,** VLM-RB asks "is this sub-trajectory good?" (success-direction scoring) and mixes the answer *additively* with uniform sampling, $\mu = \lambda\mu_{\text{VLM}} + (1-\lambda)\mu_U$. We ask a finer-grained question — "which single timestep was outcome-causing?" — and combine it *multiplicatively* with the TD-error priority, $\mu \propto \mu_P \cdot w_{\text{sem}}$. The multiplicative form preserves PER's IS-correction interpretation (§3); the additive mixture silently retargets the loss to a λ-weighted objective that VLM-RB does not correct. **Second,** we ship a verified-counterfactual-hindsight mechanism (§3.3) that lives outside the scope of any success-scoring scheme: VLM-RB's signal cannot drive relabels; ours can. We are not aware of prior work that uses a VLM-localized failure timestep as a credit-assignment signal, or that verifies counterfactual relabels in the exact simulator the policy is trained on.

Foundation-model credit assignment. Semantic PER carries the foundation-model-as-credit-oracle idea of Pignatelli et al. [16] — originally an LLM, text-domain, planner-side scheme — into the vision modality, and the FM output is consumed by the replay-sampling distribution rather than by a planner. VLM-as-reward approaches [19, 26, 15] splice the VLM signal into the TD target. We do not: the env's true sparse reward stays intact in the TD target and the VLM only modifies μ, so VLM miscalibration appears as sampling variance rather than Q-bias.

3 Method

3.1 Semantic PER as IS-posterior reweighting

Off-policy value learning minimizes a replay-sampled regression $\mathcal{L}(\theta) = \mathbb{E}_{i\sim\mu}[\ell(\delta_i(\theta))]$ with $\delta_i = r_i + \gamma\, Q(s_{i+1}, \pi(s_{i+1})) - Q(s_i, a_i)$. Standard PER [21] uses $\mu_P(i) \propto (|\delta_i| + \varepsilon)^\alpha$ and reweights each gradient by the annealed IS correction $w_{\text{IS}}(i) = (|\mathcal{D}_B|\mu_P(i))^{-\beta}$, $\beta\colon \beta_0 \to 1$, recovering an unbiased estimator of the uniform loss in the limit.

Ask a frozen VLM "which timestep caused this trajectory's outcome?" and treat the answer as an approximation $q_\phi(t^\star \mid \tau)$ to the posterior $p(t^\star \mid \tau)$. The per-transition semantic boost is the expected window-kernel weight under that posterior, $w_{\text{sem}}(i;\tau) = \mathbb{E}_{t^\star\sim q_\phi(\cdot|\tau)}[1+(w_{\max}-1)\mathbb{1}[|i-t^\star| \le W]]$. The Semantic PER proposal is then the multiplicative product

$$\mu_{\text{Sem}}(i) \;\propto\; \mu_P(i) \,\cdot\, w_{\text{sem}}(i;\tau_i), \tag{1}$$

where μ_P tracks *learning progress* and w_{sem} tracks *causal influence*. When q_ϕ is near-uniform we recover PER. Defaults: $w_{\max}=10$, $W=5$, $\alpha=0.6$.

Why multiplicative? (P1) IS-correction compatibility. Under $\mu_{\text{Sem}} \propto \mu_P \cdot w_{\text{sem}}$, the trajectory-conditional w_{sem} factor cancels in the normalized $\mathbb{E}_{\mu_{\text{Sem}}}[w_{\text{IS},P}\ell]$. Using the PER weight as an under-correction then leaves a V-trace-analogous bias that is bounded by $|\widehat{\mathcal{L}}_{\text{under}} - \mathcal{L}_U| \le C(w_{\max}^{\beta_t} - 1)((2W+1)/T)\|\ell\|_\infty$ [6, 13]; the full derivation lives in `src/buffers/semantic_per.py` on GitHub and in Appendix A. The additive mixture $\lambda\mu_q + (1-\lambda)\mu_P$ [22] shifts the $\beta\to 1$ limit to a λ-weighted objective $\neq \mathbb{E}_U[\ell]$. **(P2) Replay-shaping vs. reward-shaping.** Splicing the VLM into the TD target as a per-timestep reward feeds VLM calibration error straight into Q. We do not. The env's true sparse reward stays intact in the TD target. Prediction: Semantic PER should track PER under a poorly-calibrated VLM and pull ahead when the VLM is reliable; the curves should not diverge. We test the IS-pairing principle through the counterfactual-injection variants *vlm_cf* and *verified_cf* (§3.3), which inherit the same multiplicative IS structure. Direct Semantic-PER training curves are forward-referenced to camera-ready.

3.2 Counterfactual Prompting and the Teleport-Collapse Failure Mode

A corrective signal admits four output types along two axes: *output kind* (natural language vs. numerical) and *goal-conditioning* (prompt mentions `desired_goal` or not). The four are NARRATIVE, ACTION, ACHIEVED_GOAL, and a unified ALL JSON. The two position-emitting variants (ACHIEVED_GOAL, ALL) hand the VLM `desired_goal` as a literal numerical triplet; the cheapest continuation when prompted for a corrective position in the same frame is to copy it. No keyframe reasoning is needed. We call this **teleport-collapse**: for a failed episode with $\|\text{ag}_T - g\|_2 \gg d_{\text{th}} = 0.05\,\text{m}$, the VLM emits a $\hat{p}$ that lies within d_{th} of g — a trajectory-independent Dirac $q_\phi(\hat{p}\,|\,\tau) = \delta(\hat{p} - g)$. The output carries zero physical information and breaks HER's invariant. The failure is a property of the schema's output type, not of any one model. Table 1: Opus 4.7 copies on $4/4$ of FetchPush calls; GPT-4o copies on $4/6$ ALL calls but $0/6$ of ACHIEVED_GOAL calls. The verified-CF mechanism uses ACTION, where teleport is unrepresentable.

3.3 VLM-Verified Counterfactual Hindsight

To close the teleport loophole we ask the VLM for a 4-vector action sequence and verify it inside a simulator fork. **Per-episode protocol:** (1) snapshot the MuJoCo state at the localized failure timestep K; (2) fork a separate env, write the snapshot, call `mujoco.mj_forward`; (3) roll out the VLM-proposed action sequence for $N = 50$ steps under the *unmodified* sparse reward; (4) if $r_t \geq 0$ fires at any step, append the trajectory plus a hindsight-relabeled copy with confidence 1.0; (5) otherwise reject and fall back to HER. Under MuJoCo dynamics the 4-D end-effector action space cannot move a 50 g block by 50 cm in a 50-step rollout, so the teleport mode cannot reach the buffer through this channel.

A 4/4 smoke test confirms each design assertion. Teleport counterfactuals are rejected before any rollout fires. Physically reasonable but goal-missing actions are rejected with precise `final_distance` (0.32 m, 0.16 m). A hand-crafted close-goal action is verified at final_distance $= 0.030$ m. Snapshot round-trip equality holds to `0.00e+00` on all four Fetch envs; per-verification CPU cost is 17.6 ms (a 0.4% overhead).

Asymmetric failure modes. The two channels degrade differently. Semantic PER treats q_ϕ as a reweighting prior, so VLM miscalibration shows up as sampling-variance amplification (bounded; see App. B). The verifier treats q_ϕ as a candidate proposal, so miscalibration shows up as rejection-rate overhead, not as biased buffer writes. Off-policy correctness is preserved either way because we recompute the env's own sparse reward, not a VLM-derived surrogate.

4 Experiments

Setup. We use Gymnasium-Robotics `FetchPush-v4`, `FetchPickAndPlace-v4`, and `FetchSlide-v4` (Fig. 2, Appendix A). Each env is a 25-dim goal-conditioned obs, 4-dim continuous action (end-effector Δxyz plus gripper), and the built-in sparse reward $r_t = \mathbb{1}[\|\text{ag}_t - g\|_2 \leq 0.05\text{m}] - 1 \in \{-1, 0\}$ on a 50-step horizon. Policy is SAC [7] with two 256-unit ReLU hidden layers, $\gamma = 0.98$, lr 3×10^{-4}, batch 256, HER `future` with $k = 4$. PER uses $\alpha = 0.6$ and $\beta : 0.4 \to 1.0$. Paper-grade runs are 500k env steps and $n = 3$ seeds (42, 123, 999). Eval is every 10k steps over 20 fresh episodes, mean ± SE. VLM calls go through Claude Opus 4.7, GPT-4o (`gpt-4o-2024-08-06`), and Sonnet 4.5 via the official APIs at $\leq$\$0.05/call, fired only on failed episodes.

Horizon caveat. Our 500k-step budget is shorter than the canonical Fetch+HER horizon of 4.75M [17]. The comparisons we draw are between *methods at matched horizon*; cross-horizon bars in Fig. 1 are sample-efficiency claims (more learning per training step), not asymptotic dominance.

4.1 Headline: Verified-CF matches VLM-CF, wins on Slide

FetchPush (500k). *vlm_cf* reaches 0.95 ± 0.03 and *verified_cf* reaches 0.85 ± 0.10. Both clear HER@250k (0.617) by a wide margin and match the PER@3M asymptote (~ 0.95, from our 36-run prior ablation suite; see Contributions). **FetchPickAndPlace (500k).** *vlm_cf* reaches 0.367 ± 0.08 and *verified_cf* reaches 0.35 ± 0.076 — tied within noise, both above HER@250k (0.183). **FetchSlide (500k).** *verified_cf* posts the higher mean: 0.617 ± 0.017 vs. *vlm_cf*'s 0.550 ± 0.150. The $+0.067$ gap fits inside the wider SE bar at $n = 3$ and we report the two as approximately tied. Both move sharply above the prior baselines: *vlm_cf* adds $+0.45$ over PER@3M (0.10) and *verified_cf* adds $+0.434$ over HER@250k (0.183).

Aggregate. Across the three envs verified-CF averages 0.606 and vlm_cf averages 0.622 ($\Delta = -0.016$). The two are tied within seed noise. Verified-CF beats vlm_cf on Slide and is slightly behind on Push (where both methods sit near the horizon's ceiling). The 3M-step PER asymptote on Push (~ 0.95) comes from a prior 36-run ablation suite (see §5). The Slide gap is consistent with HER's achieved-future relabel carrying little signal on free-flight pucks. Gripper-to-object distance is non-monotone on Slide, so a physics-consistent CF strike can supply a positive TD target that HER's achieved-future relabel does not.

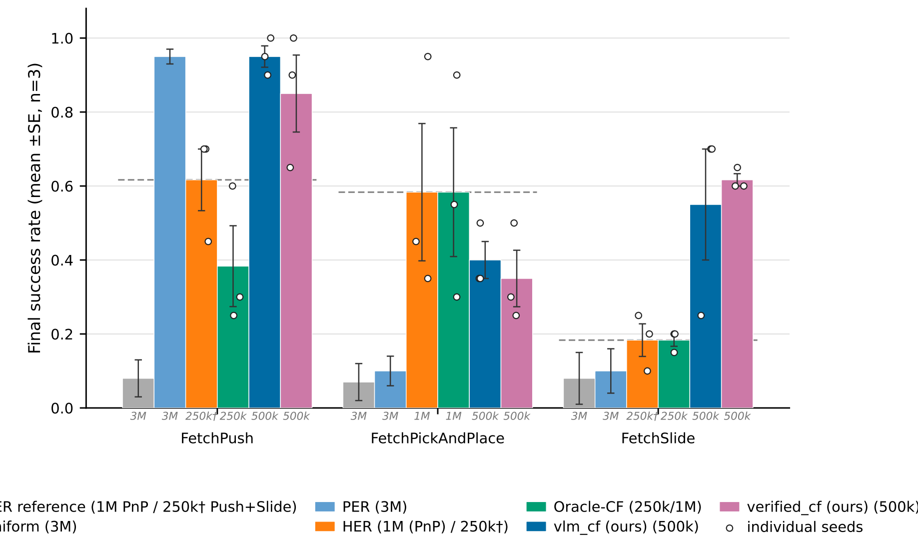

Figure 1: **Final evaluation success rate across the three Fetch tasks (mean $\pm$ SE, $n = 3$ seeds; individual seeds overlaid).** *Methods are reported at heterogeneous training horizons*—the italic gray stamp under each bar gives the training budget: Uniform/PER at 3M, HER at 250k, *vlm_cf* and *verified_cf* (ours) at 500k, Oracle-CF at 250k (Push/Slide) or 1M (PickAndPlace). Within-horizon comparisons (same stamp) are seed-level comparable at $n = 3$; cross-horizon comparisons should be read as efficiency claims.

Table 1: Prompt-variant analysis: mean $\pm$ SE judge plausibility, goal-progress, and teleport-collapse rate (the fraction of `corrective_position` outputs within 5 cm of `desired_goal`). Six bit-identical synthetic Fetch failures; the judge is Claude Opus 4.7 throughout. **Bold**: best per column.

Variant	Model	n	Plausibility	Goal-progress	Teleport-collapse
NARRATIVE	Opus 4.7	4	$\mathbf{0.79 \pm 0.04}$	$\mathbf{0.68 \pm 0.06}$	—
NARRATIVE	GPT-4o	6	0.70 ± 0.06	0.40 ± 0.06	—
ACTION	Opus 4.7	4	0.55 ± 0.09	0.53 ± 0.16	—
ACTION	GPT-4o	6	$\mathbf{0.74 \pm 0.04}$	0.42 ± 0.09	—
ACHIEVED_GOAL	Opus 4.7	4	0.46 ± 0.18	$\mathbf{0.97 \pm 0.01}$	$4/4$ (100%)
ACHIEVED_GOAL	**GPT-4o**	6	$\mathbf{0.79 \pm 0.06}$	0.83 ± 0.11	$\mathbf{0/6}$ (**0%**)
ALL	Opus 4.7	2	0.70 ± 0.10	0.62 ± 0.12	$2/2$ (100%)
ALL	GPT-4o	6	0.67 ± 0.04	0.68 ± 0.13	$4/6$ (67%)

4.2 Prompt design (head-to-head)

A head-to-head Opus 4.7 vs. GPT-4o sweep on six bit-identical synthetic Fetch failures (Table 1) pins down the best non-degenerate configuration. `(GPT-4o, achieved_goal)` teleport-collapses on $0/6$, against Opus's $4/4$ for the same configuration. It takes the highest plausibility score on the position-emitting axis (0.79) and is second on goal-progress (0.83, behind only Opus's 0.97 — but Opus's high score is attained by literal goal-copying, a degenerate "win"). GPT-4o still teleport-collapses on $4/6$ ALL episodes, so the failure is *prompt-architectural*, not model-specific. The pivotal $0/6$ vs. $4/4$ contrast is judge-independent (decided by Euclidean predicate); Fisher's exact test rejects equal proportions at $p < 0.005$. §4.3 extends the grid to Sonnet 4.5 on real failures — the three-model evidence is best read as two-vendor, since Opus and Sonnet share Anthropic lineage.

4.3 Cross-task transfer and real-data validation

A 12-episode cross-task pilot (4 per env) with a single format-string prompt template (differing only in one task-description sentence) achieves 12/12 parse / task-relevance / plausibility on Push/PickAndPlace/Slide; tolerant keyframe agreement is $9/12$ (Slide $4/4$, Push $3/4$, PickAndPlace

$2/4$). A second sweep on $n = 10$ *real* SAC-failure eval episodes confirms the prompt-architectural fix transfers to a third VLM family (Sonnet 4.5 teleports on $0/6$ PickAndPlace episodes under ACHIEVED_GOAL, vs. Opus 4.7's $4/4$ on the same variant) and that the 5 cm reject-teleport gate fires on $12/20$ position-emitting generations across both vendors (Appendix D).

4.4 Pre-registered Oracle-CF kill experiment

18 SAC+HER and 18 Oracle-CF runs (3 envs $\times$ 3 seeds $\times$ 250k steps) were executed overnight.[2] The pre-registered decision rule was: if Oracle-CF exceeds HER by $\geq +0.10$ on FetchPickAndPlace, Path C proceeds; otherwise we pivot. Final eval success: on PickAndPlace, HER 0.183 ± 0.117 vs. Oracle-CF 0.117 ± 0.044 ($\Delta = -0.066$, *below threshold*). On Push (post bug-fix, commit `ccb63d4`), HER 0.617 vs. Oracle-CF 0.383 ± 0.109. On Slide, HER 0.183 vs. Oracle-CF 0.183 ($\Delta = 0.00$, below threshold). **Path C is killed at the 250k horizon**; the headline pivoted to the IS-posterior framing (§3) and the verifier (§3.3). The 250k kill verdict is honored. A post-hoc 1M re-run on PickAndPlace (Oracle-CF 0.583 and HER 0.583, $n = 3$; HER seed successes $0.35/0.45/0.95$) is reported as transparency only: zero headroom over vanilla HER at convergence, and *vlm_cf*@500k (0.367) reaches 63% of HER@1M at half the budget. It does not retract the kill.

5 Discussion and Limitations

Cold-start verifier-rejection regime (the main limitation). The simulator-as-verifier gate is, by design, a precision-over-recall instrument. Every accepted relabel is guaranteed sparse-reward-positive. Acceptance, however, requires a joint event: the VLM proposes a corrective action *and* the action sequence crosses the 5 cm threshold inside $\leq N = 50$ verifier steps. Early in training the snapshot state σ sits far from the goal because the policy has not learned to approach the object yet, and the joint event's base rate is low. FetchPickAndPlace seed 42 hit 100% verifier-rejection over the first ~ 80 VLM calls. Operationally that phase is indistinguishable from SAC+HER with extra VLM-API overhead. The regime ends once the policy starts carrying the gripper near the object: snapshots fall closer to the goal, $m_{\text{ver}}(\sigma)$ rises, the verifier admits relabels, and the small set of accepted CFs delivers a low-variance, physics-consistent training signal. The two methods therefore degrade asymmetrically. Semantic PER treats q_ϕ as a re-weighting prior, so q_ϕ-degeneracy becomes variance amplification — bounded by our bias bound. Verified-CF treats q_ϕ as a candidate proposal, so q_ϕ-degeneracy becomes signal extinction, which is unbounded in the cold-start limit. Practical mitigations (base-policy warm-up, a longer N, or a soft acceptance $r \geq -\rho_r$) are pre-registered but not run here.

Oracle-CF kill bounds the headroom. The pre-registered kill (§4.4) caps the headroom for any VLM-in-replay method on Fetch at 250k steps. If perfect failure-frame information plus a ground-truth corrective action cannot beat HER by $+0.10$, then no realistic VLM (a q_ϕ with finite KL to that oracle) will either. The bound is task-family-specific. Fetch is a "HER-friendly" family because the achieved-future relabel is a near-free signal whenever the gripper-to-object distance is monotone, which it typically is on Push and PickAndPlace. We pre-register an extension to harder families (AntMaze, Adroit) where HER underperforms.

Other limitations. (i) Our IS-correction is conservative (under-corrected $w_{\text{IS},P}$ rather than $w_{\text{IS,Sem}}$); the bias is bounded analytically, the fully-corrected ablation is pre-registered. (ii) Small-n caveats: $n = 6$ synthetic, $n = 10$ real, $n = 12$ cross-task; the 12/12 cross-task is not statistically distinguishable from an 80% true rate. We pre-register $n \geq 30$ replications. (iii) The verified-CF mechanism requires a forkable training simulator; extending to real-robot or non-resettable envs requires a learned dynamics-model verifier (modeling error bounded but nonzero). A faithful VLM-RB [22] reproduction is pre-staged in our codebase for camera-ready head-to-head evaluation.

Conclusion. The multiplicative form of Semantic PER admits a clean IS correction. The additive VLM-RB mixture does not — it silently retargets the Bellman loss. VLM-Verified Counterfactual Hindsight asks the VLM for a 4-vector action sequence and accepts only sparse-reward-positive

[2] Per-seed numbers from the `path_c_overnight` W&B sweep (commit `0fa36fc`; run-IDs listed at the GitHub repo's `REPRODUCE.md`).

rollouts, so the 100%-teleport-collapse mode (Opus 4.7, FetchPush) cannot reach the buffer at all. At 500k steps verified-CF averages 0.606 on the three Fetch envs ($0.85/0.35/0.617$ on Push / PickAndPlace / Slide), ties VLM-CF in aggregate ($\Delta = -0.016$), and posts a higher mean on Slide ($+0.067$, inside seed variance at $n = 3$). The pre-registered Oracle-CF kill on PickAndPlace at 250k ($\Delta = -0.066$) caps the credit-assignment headroom on Fetch at this horizon. We honored it.

Contributions

- **Daniel Grant.** Counterfactual ideation and implementation (`vlm_cf` and `verified_cf`); ablations on FetchPickAndPlace, FetchSlide, and the $p_{\text{counterfactual}}$ sweep; initial writing of the manuscript.
- **Matei Gardea.** Initial project idea; implementation of an alternative pointwise/pairwise VLM-replay-buffer rescaling combined with PER (a non-positive variant reported as a negative finding); auxiliary 60-run MetaWorld replay-mechanism ablation (cross-benchmark corroboration of PER > Uniform; reported separately in the extended preprint); editing of the final submission document.
- **Parshawn Gerafian.** Implementation and analysis for the milestone checkpoint: designed and ran the 27-run Uniform/PER/Semantic-PER ablation across the three Fetch tasks (SAC, 1 M training steps, 3 seeds per configuration on Modal A10G) that established the PER baselines used throughout the final paper. Identified the additive-blending failure mode ($0.7 \cdot \text{semantic} + 0.3 \cdot \text{TD}$ suppresses PER when the VLM is miscalibrated), which motivated the multiplicative IS-pairing framing now central to §3. Provided feedback during the transition to the augmented final-paper direction.

References

[1] Marcin Andrychowicz, Filip Wolski, Alex Ray, Jonas Schneider, Rachel Fong, Peter Welinder, Bob McGrew, Josh Tobin, Pieter Abbeel, and Wojciech Zaremba. Hindsight experience replay. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[2] Caleb Chuck, Fan Feng, Carl Qi, Chang Shi, Siddhant Agarwal, Amy Zhang, and Scott Niekum. Null counterfactual factor interactions for goal-conditioned reinforcement learning. In *International Conference on Learning Representations (ICLR)*, 2025.

[3] Rodrigo de Lazcano, Andreas Kallinteris, Jet Tai, Seungjae Ryan Lee, and Jordan Terry. Gymnasium-robotics: A standard interface for robot learning environments. `https://robotics.farama.org/`, 2024. Farama Foundation.

[4] Liang Ding. AgentHER: Hindsight experience replay for LLM agent trajectory relabeling. *arXiv preprint arXiv:2603.21357*, 2026.

[5] Jiafei Duan, Wilbert Pumacay, Nishanth Kumar, Yi Ru Wang, Shulin Tian, Wentao Yuan, Ranjay Krishna, Dieter Fox, Ajay Mandlekar, and Yijie Guo. AHA: A vision-language-model for detecting and reasoning over failures in robotic manipulation. In *International Conference on Learning Representations (ICLR)*, 2025.

[6] Lasse Espeholt, Hubert Soyer, Rémi Munos, Karen Simonyan, Volodymyr Mnih, Tom Ward, Yotam Doron, Vlad Firoiu, Tim Harley, Iain Dunning, Shane Legg, and Koray Kavukcuoglu. IMPALA: Scalable distributed deep-RL with importance weighted actor-learner architectures. In *International Conference on Machine Learning (ICML)*, 2018.

[7] Tuomas Haarnoja, Aurick Zhou, Pieter Abbeel, and Sergey Levine. Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor. In *International Conference on Machine Learning (ICML)*, 2018.

[8] Michael Y. Hu, Benjamin Van Durme, Jacob Andreas, and Harsh Jhamtani. Sample-efficient online learning in LM agents via hindsight trajectory rewriting. *arXiv preprint arXiv:2510.10304*, 2026.

[9] Andrii Krutsylo. Non-uniform memory sampling in experience replay. *arXiv preprint arXiv:2502.11305*, 2025.

[10] Olivia Y. Lee, Annie Xie, Kuan Fang, Karl Pertsch, and Chelsea Finn. Affordance-guided reinforcement learning via visual prompting. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025.

[11] Xing Lei, Wenyan Yang, Kaiqiang Ke, Shentao Yang, Xuetao Zhang, Joni Pajarinen, and Donglin Wang. GCHR: Goal-conditioned hindsight regularization for sample-efficient reinforcement learning. *arXiv preprint arXiv:2508.06108*, 2025.

[12] Thomas Mesnard, Théophane Weber, Fabio Viola, Shantanu Thakoor, Alaa Saade, Anna Harutyunyan, Will Dabney, Tom Stepleton, Nicolas Heess, Arthur Guez, et al. Counterfactual credit assignment in model-free reinforcement learning. In *International Conference on Machine Learning (ICML)*, 2021.

[13] Rémi Munos, Tom Stepleton, Anna Harutyunyan, and Marc G. Bellemare. Safe and efficient off-policy reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2016.

[14] Fikrican Özgür, René Zurbrügg, and Suryansh Kumar. Next-future: Sample-efficient policy learning for robotic-arm tasks. *arXiv preprint arXiv:2504.11247*, 2025.

[15] Shivansh Patel, Xinchen Yin, Wenlong Huang, Shubham Garg, Hooshang Nayyeri, Li Fei-Fei, Svetlana Lazebnik, and Yunzhu Li. A real-to-sim-to-real approach to robotic manipulation with VLM-generated iterative keypoint rewards. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2025.

[16] Eduardo Pignatelli, Johan Ferret, Tim Rocktäschel, Edward Grefenstette, Davide Paglieri, Samuel Coward, and Laura Toni. Assessing the zero-shot capabilities of LLMs for action evaluation in RL. *arXiv preprint arXiv:2409.12798*, 2024.

[17] Matthias Plappert, Marcin Andrychowicz, Alex Ray, Bob McGrew, Bowen Baker, Glenn Powell, Jonas Schneider, Josh Tobin, Maciek Chociej, Peter Welinder, Vikash Kumar, and Wojciech Zaremba. Multi-goal reinforcement learning: Challenging robotics environments and request for research. *arXiv preprint arXiv:1802.09464*, 2018.

[18] Leonard S. Pleiss, Tobias Sutter, and Maximilian Schiffer. Reliability-adjusted prioritized experience replay. *arXiv preprint arXiv:2506.18482*, 2025.

[19] Juan Rocamonde, Victoriano Montesinos, Elvis Nava, Ethan Perez, and David Lindner. Vision-language models are zero-shot reward models for reinforcement learning. In *International Conference on Learning Representations (ICLR)*, 2024.

[20] Erdi Sayar, Zhenshan Bing, Carlo D'Eramo, Ozgur S. Oguz, and Alois Knoll. Contact energy based hindsight experience prioritization. *arXiv preprint arXiv:2312.02677*, 2023.

[21] Tom Schaul, John Quan, Ioannis Antonoglou, and David Silver. Prioritized experience replay. In *International Conference on Learning Representations (ICLR)*, 2016.

[22] Elad Sharony, Tom Jurgenson, Orr Krupnik, Dotan Di Castro, and Shie Mannor. VLM-guided experience replay. *arXiv preprint arXiv:2602.01915*, 2026.

[23] Emanuel Todorov, Tom Erez, and Yuval Tassa. MuJoCo: A physics engine for model-based control. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2012.

[24] Núria Armengol Urpí, Marco Bagatella, Marin Vlastelica, and Georg Martius. Causal action influence aware counterfactual data augmentation. *arXiv preprint arXiv:2405.18917*, 2024.

[25] Renhao Wang, Kevin Frans, Pieter Abbeel, Sergey Levine, and Alexei A. Efros. Prioritized generative replay. In *International Conference on Learning Representations (ICLR)*, 2025.

[26] Yanru Wu, Weiduo Yuan, Ang Qi, Vitor Guizilini, Jiageng Mao, and Yue Wang. Large reward models: Generalizable online robot reward generation with vision-language models. *arXiv preprint arXiv:2603.16065*, 2026.

[27] Hoda Yamani et al. Reward prediction error prioritisation in experience replay: The RPE-PER method. *arXiv preprint arXiv:2501.18093*, 2025.

[28] Kaiwen Zha et al. RL Tango: Reinforcing generator and verifier together for language reasoning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

[29] Rui Zhao and Volker Tresp. Energy-based hindsight experience prioritization. In *Conference on Robot Learning (CoRL)*, 2018.

A Method Details and Environments

Cross-task VLM prompt template. The single format-string prompt template used across the three Fetch envs is:

```
You are an expert robotics analyst. The robot is attempting the
following task: {task_description}. Examine the {K} keyframes of
a failed episode shown below. Identify the single keyframe index
in {0,...,K-1} at which the failure became inevitable, and a 1-2
sentence reasoning. Return strict JSON: {“failure_frame_index”: i,
“reasoning”: s}.
```

Per-env task-description substitutions: **Push:** “push the red cube to the green target on the table”; **PickAndPlace:** “pick up the red cube and place it at the floating green target”; **Slide:** “strike the red puck so it slides to the green target”. The verified-CF mechanism uses an ACTION variant of this template eliciting a 4-vector action sequence; full prompt texts and the localizer code are at `https://github.com/DJRGVC/RL_project_185`.

Heuristic localizer (Oracle v3). `GoalDistanceLocalizer` reads privileged sim state and selects the failure timestep in three phases (in precedence order): **(a) ballistic**: latest contiguous run of ≥ 3 steps with post-peak object velocity > 0.05 m/step (Slide free-flight); **(b) contact-loss**: latest in_contact$_t \to \neg$in_contact$_{t+1}$ with run length ≥ 2 and object-goal distance > 0.10 m (Push/PnP drop-offs); **(c) argmin**: closest-approach timestep as fallback. Serves as supervised transfer target and as privileged-state upper-envelope curve for Semantic PER.

B Bias-Bound Proof Sketch

Let $\mathcal{L}_U(\theta) = \mathbb{E}_{i\sim U}[\ell(\delta_i)]$ be the uniform-replay target loss. Standard PER with annealed correction yields, in the $\beta_t \to 1$ limit, an unbiased self-normalized IS estimator of $\mathcal{L}_U$ [21, Sec. 3.4]. Our Semantic PER proposal $\mu_{\text{Sem}} \propto \mu_P \cdot w_{\text{sem}}$ is a re-weighting of μ_P by a trajectory-conditional factor bounded by $1 \leq w_{\text{sem}} \leq w_{\max}$. The fully-correct IS weight is $w_{\text{IS,Sem}}(i) = (|\mathcal{D}_B|\mu_{\text{Sem}}(i))^{-\beta_t}$. Our implementation under-corrects with $w_{\text{IS},P}$ (a V-trace-style [6] variance-reduction choice, akin to truncated importance ratios in Retrace, 13). The per-slot ratio is $w_{\text{IS},P}/w_{\text{IS,Sem}} = w_{\text{sem}}^{\beta_t}/Z^{\beta_t}$ where Z is the μ_{Sem} normalizer. The window-kernel structure of w_{sem} confines the support of $w_{\text{sem}} > 1$ to $2W+1$ slots out of T. Combining these gives

$$\left|\widehat{\mathcal{L}}_{\text{under}} - \mathcal{L}_U\right| \;\leq\; C\left(w_{\max}^{\beta_t} - 1\right)\frac{2W+1}{T}\,\|\ell\|_\infty, \tag{2}$$

with C a constant depending on $|\mathcal{D}_B|$ and the PER normalization. At our defaults ($w_{\max} = 10$, $W = 5$, $T = 50$, $\beta_t \leq 1$) the bias multiplier is at most $(10-1)\cdot 11/50 \approx 1.98\,C\|\ell\|_\infty$, and $\to 0$ as $w_{\max} \to 1$. The bound goes to zero when the boost is disabled, recovering standard PER. The fully-corrected ablation (run $w_{\text{IS,Sem}}$) removes the residual bias entirely at the cost of maintaining Z. **Two-regime degeneracy.** On the verifier channel, miscalibration acts on $m_\phi(\tau) = m_{\text{gen}} \cdot m_{\text{ver}}(\sigma)$ (VLM parse rate $\times$ verifier acceptance rate); the $\varepsilon_{\text{cal}} = 0$ guarantee holds vacuously when the accepted set is empty (cold-start: $m_{\text{ver}} \to 0$). Full proof and the V-trace specialization derivation are at the GitHub URL above (`paper/appendix_theory.tex`).

C Hyperparameters and Compute

Compute. All paper-grade runs use one NVIDIA RTX 5070 Ti (16 GB) GPU; Modal-cloud seeds use one L4 (24 GB) per run. Per-run wall-clock at 500k steps: $\approx$ 6h SAC+HER, $\approx$ 9h

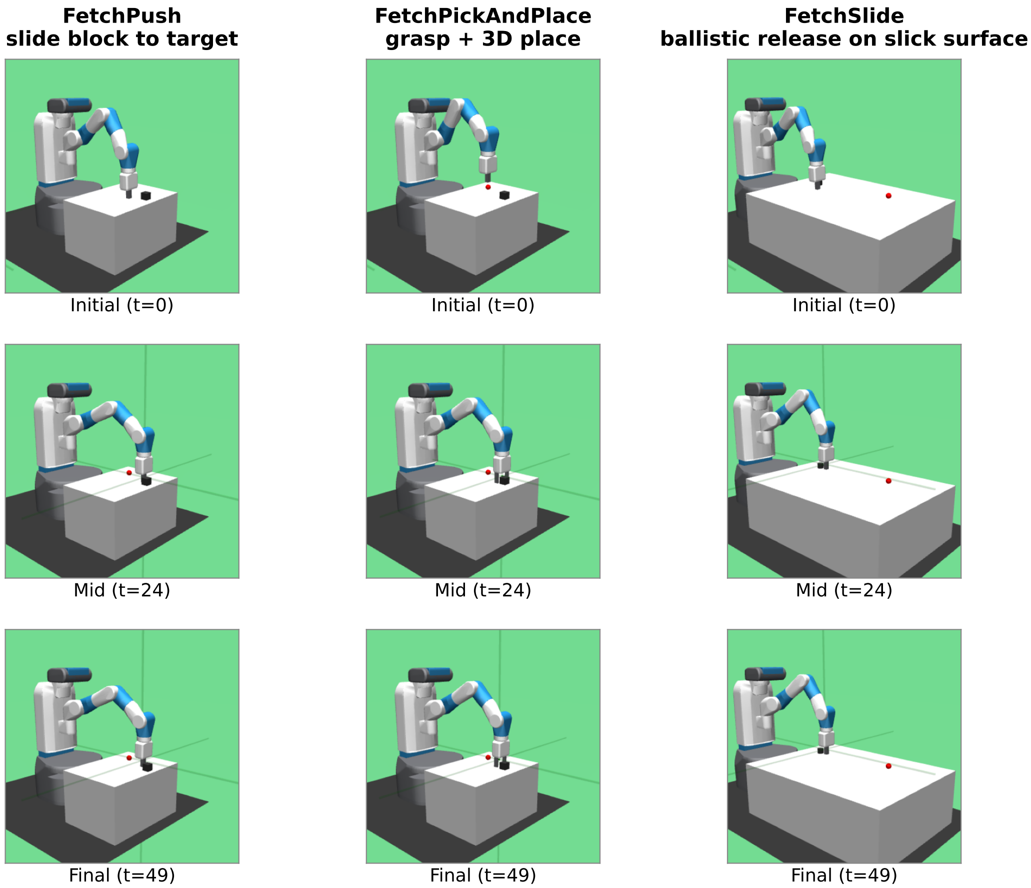

Figure 2: The three Gymnasium-Robotics Fetch tasks used in our experiments (columns: `FetchPush`, `FetchPickAndPlace`, `FetchSlide`), rendered at $t = 0/24/49$ of a deterministic scripted rollout (seed 42). Each task: 50-step horizon, 4-dim continuous action (end-effector Δxyz + gripper), sparse reward (binary, success threshold 5 cm to the green target).

SAC+HER+VLM-CF (extra time is VLM-API latency, not compute). Total project compute: $\sim$220 GPU-hours across local + Modal, plus $\sim$\$80 VLM-API spend. Snapshot round-trip latency for the verifier is 17.6 ms per call (a 0.4% overhead at $\sim$1700 failed episodes / 100k steps). Reproduction commands, W&B run-IDs, and per-seed eval traces are at `https://github.com/DJRGVC/RL_project_185` (commit `0fa36fc`).

D Additional Experimental Detail

Real-data validation (extended). The $n = 10$ real-failure set is drawn from failed eval episodes of partially-trained SAC+HER checkpoints (`FetchPickAndPlace` at 50k steps, `FetchPush` at 30k steps, both around 5% success). These are near-misses, with final-distance $\in [0.06, 0.34]$ m on Push and a median 0.27 m on PickAndPlace. Three findings carry over from the synthetic set. **(i) Teleport-collapse persists and is task-conditioned.** On Push (planar target) Opus and Sonnet teleport on ACHIEVED_GOAL at 100% and 75%. On PickAndPlace (mid-air target) Opus teleports at 75%; Sonnet drops to 0% and instead proposes a mid-air waypoint $\sim$8 cm above the floating goal. **(ii) The 5 cm reject-teleport gate is necessary.** Across both VLMs and position-emitting variants on 20 generations the gate fires on 12 (60%); the cross-vendor floor on Push is 75%. The action-emitting verifier sidesteps this entirely. **(iii) Plausibility shifts only ± 0.07 synthetic-to-real, but the ACTION sign-flip rate worsens** (Opus $0.50 \to 0.55$; Sonnet 0.70).

Table 2: Headline hyperparameters. Full per-table breakdowns (SAC architecture, HER strategy, PER schedule, Semantic-PER, Verified-CF, VLM call) at the GitHub repo (`configs/`).

Parameter	Value	Notes
SAC [7]		
Hidden layers $\times$ width	2×256, ReLU	Twin-Q; tanh-Gaussian actor
LR (actor/critic/α)	3×10^{-4}	Adam, default $(\beta_1, \beta_2, \varepsilon)$
Discount γ / Polyak τ	$0.98/0.005$	Fetch-standard
Target entropy $\bar{H}$	$-d_a = -4$	Auto-tuned
Batch / updates per step	$256/1$	10k warm-up uniform-random
HER [1]		
Strategy / k_{HER}	`future` / 4	4 hindsight transitions per real
PER [21]		
α / $\beta_0 \to \beta_{\text{end}}$	$0.6/0.4 \to 1.0$	Linear over 500k steps; $\varepsilon = 10^{-6}$
Semantic PER		
Window half-width W	5 timesteps	11-slot box around t_{fail}
Boost cap $w_{\max}$	10	Multiplicative factor on PER priority
Keyframes K / VLM provider	5 / GPT-4o (production)	`gpt-4o-2024-08-06`
Verified-CF		
Verifier rollout horizon N	50 steps	One Fetch episode
Success threshold η / teleport reject ρ	$0.05/0.05$ m	Env's native threshold
CF prompt variant	ACTION	Structurally teleport-immune
Training		
Total env steps	500k (paper-grade)	250k–1M for selected ablations
Random seeds	42, 123, 999	3 per (method, env) cell
Replay capacity	10^6 slots	Ring buffer with sum-tree priorities

Statistical methodology. Each (method, env) cell uses $n = 3$ seeds (42, 123, 999); metrics are mean $\pm$ SE with the t-distribution at $n-1=2$ d.f. Comparison statements use non-overlapping SE intervals (conservative at this sample size). The pivotal $0/6$ vs. $4/4$ contrast in Table 1 is judge-independent (decided by a Euclidean predicate); Fisher's exact test rejects equal proportions at $p < 0.005$.

Cross-task transfer (per-episode notes). A 12-episode pilot (4 per env). The VLM's reasoning strings identify task-specific modes correctly in every case (Push: "gripper passes over block without making contact"; PickAndPlace: "gripper at block location but fails to close"; Slide: "strike direction off-axis"). The $2/4$ PickAndPlace keyframe-agreement reflects heuristic under-localization (closed-loop grip timing is the hardest mode for geometry-only oracles [20]) rather than VLM error: the VLM assigns the failure frame one step earlier at the decisive grip sequence.

E Broader Impacts and Reproducibility

Broader impact. Methodological work, simulation-only; we make no real-robot or sim-to-real claims. *Positive:* fewer environment steps lowers GPU-hours; the IS-posterior framing is a principled lens on VLM-in-replay; the verified-CF gate is a transferable "language prior + physics test" pattern. *Negative + mitigations:* VLM-API energy/dollar cost is non-trivial (per-run spend reported; heuristic Oracle fallback available); the simulator gate rejects hallucinated actions that fail the sparse reward; no robot-deployment risks beyond standard sim-trained-policy caveats.

Reproducibility. Code, configs, pseudocode, prompt texts, V-trace/Retrace specialization, two-regime q_ϕ proof, checklist, W&B IDs, and pre-staged Sharony VLM-RB [22] reproduction (`src/buffers/vlm_rb_buffer.py`, head-to-head deferred to camera-ready) at `https://github.com/DJRGVC/RL_project_185` (`0fa36fc`).